

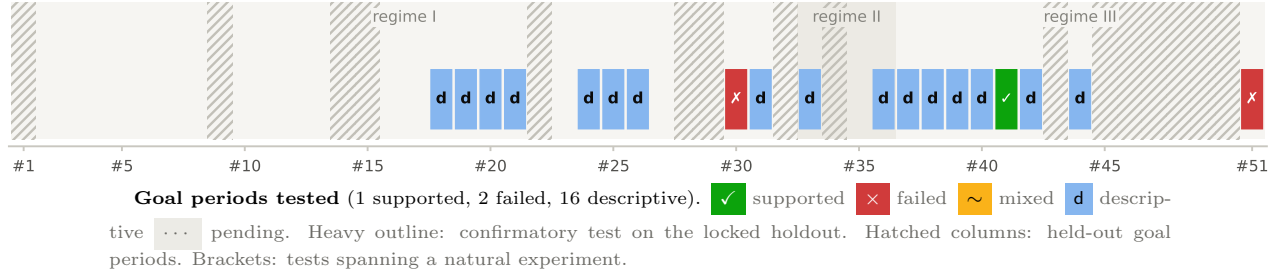
H138: Glauber escape from a project grows with the number of open options

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 1/5 Usefulness 1/5

Bottom line. Within goal periods the per-call leave hazard does not grow with the number of open projects (pooled elasticity $-0.66 [-1.64, 0.32]$ work, $-0.28 [-0.42, -0.15]$ attention), so the Glauber law $\varepsilon_q = 1$ is rejected; the HH381 kill itself is unpowered.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

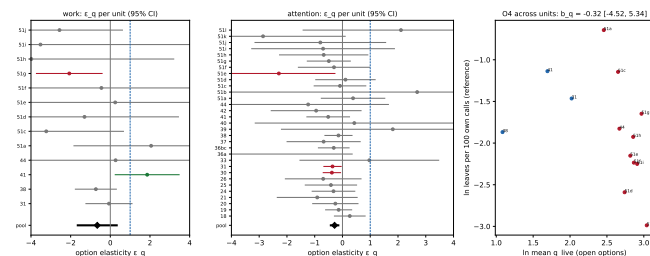
HH381 says that an agent working on one project leaves in the way a Glauber spin leaves its Potts state: at each of its own calls, the chance of leaving grows in proportion to the number of other open options, $h \propto (q-1)e^{-\beta\Delta}$. H138 builds one row per agent, 30-min window and project visit (work: DQ4 host labels; attention: strict project mentions) and fits the leave hazard per own call against the number of projects other agents held in the last 2 active hours. **Leaving does not rise with the open options:** the pooled option elasticity is $-0.66 [-1.64, 0.32]$ in the work channel (13 units) and $-0.28 [-0.42, -0.15]$ in the attention channel (31 units). The literal law $\varepsilon_q = 1$ is rejected. The card's kill ($\varepsilon_q = 0$ vs 0.5) is unpowered in the work channel (0.43), so the verdict by its rule is inconclusive.

PHYSICAL MODEL

Models 10 (kinetic Potts) and 02 (single-spin kinetics). Heat-bath walker on the agent's call clock: from project a it moves to b with probability $\nu e^{u_b} / (e^{u_a} + \sum_{b'} e^{u_{b'}})$. For rare leaves $h \approx \nu e^{-u_a} Z_{\text{alt}}$; with equal attractions $Z_{\text{alt}} = (q-1)e^u$, so $\varepsilon_q = \partial \ln h / \partial \ln(q-1) = 1$ at a fixed stay field. Rival R-finish: the agent leaves when its task ends, $\varepsilon_q \approx 0$.

$$\text{cloglog } P(\text{leave in } w) = \ln c_{iw} + \theta_i + s(t) + \varepsilon_q \ln q_{\text{live}} + \gamma_d \ln(1+d) + \beta_a \ln(1+a_a) + \beta_c m + \beta_{\text{own}o}$$

$$q_{\text{live}}(i, w) = \#\{b \neq a : \text{ held by } j \neq i \text{ in } w - 4..w - 1\}, \quad Z_{\text{alt}} = \sum_{b \in C} e^{u_b}$$



Observables. (a) Work and (b) attention option elasticity $\hat{\varepsilon}_q$ per testable unit (cloglog hazard per own call, agent effects, active-time spline; agent-cluster 95% CIs; green CI above 0, red below 0). Black: DerSimonian–Laird pool. Dashed line: the Glauber value 1. (c) Cross-unit reference leave rate against the mean number of open options (blue shared-goal, red own-role units); the slope is not identified (power 0.12).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: opening more parallel projects does not make agents drop their current one faster. Expect about 0.06–0.3 leaves per 100 own calls in the work channel whatever the number of open projects; in own-role weeks agents leave slightly less when many projects are live. Plan churn from task length, not from the size of the menu. Whether naming a target project pulls agents is open (a small attention signal, below the label-noise level).

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
S1–S3 synthetic	unbiased; size 0.055; power 0.43 at 0.5	passed
P1 $\varepsilon_q > 0$ (work)	1/13 units; −0.66 [−1.64, 0.32]	pool against
A3 bound $\varepsilon_q < 1$	upper limit 0.32	$\varepsilon_q = 1$ rejected
P2 $\varepsilon_Z \in [0.5, 1.5]$	0.22 [−0.12, 0.55]; Z wins 8/13	not passed
P3 lag > lead	−0.51 [−1.47, 0.45]	failed
P4 $R_{\text{own}} \in [0.5, 2]$	0/13	failed
P5 cross-unit b_q	−0.32 [−4.5, 5.3]	descriptive
N1 G38 terciles	ratio 2.89 vs $\times 1.5$	against
N2 G44 rooms	HR 1.97 vs 3.32 (15 leaves)	descriptive
N3 G51	−1.83 [−3.19, −0.46]	failed
Kill (work)	CI includes 0; power 0.43	untestable
Attention	−0.28 [−0.42, −0.15]; power 1.00	against

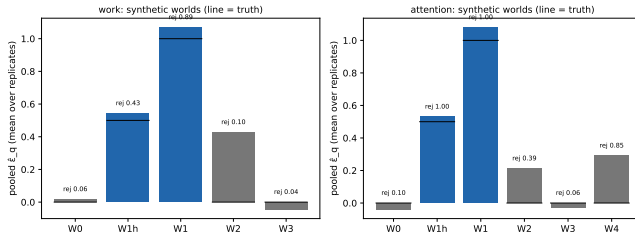
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Work leaves are few outside #51 (25–88 per unit), so no single work unit has power 0.8. The attention negative appears only after the active-time spline (agent-only fit +0.18 [−0.02, 0.38]; post hoc). The #51 negative rests on Wald CIs; the within-day permutation gives $p \geq 0.095$ in each #51 unit. Labels are 30-min windows, coarser than read-out delays. Reserved periods were not run.



Second figure. Synthetic validation on the real skeleton of every testable unit (200 replicates per world): pooled $\hat{\varepsilon}_q$ by world (line: truth) with its rejection rate of $\varepsilon_q = 0$. The estimator is unbiased; renewal (W0) and shared-drive (W3) worlds keep size 0.04–0.06. Work power is 0.89 at $\varepsilon_q = 1$ but 0.43 at 0.5. Attention label noise (W4) fakes $\hat{\varepsilon}_q \approx 0.3$.

GOAL PERIODS IN DETAIL

G18 d descriptive
G19 d descriptive
G20 d descriptive
G21 d descriptive
G24 d descriptive
G25 d descriptive
G26 d descriptive
G30 × failed
G31 d descriptive
G33 d descriptive
+9 more periods: see the card.

ROUND 2

Refit on the call clock (one row per own call) to raise power and make the lead placebo work; explain the negative elasticity with a same-time cross-agent q shuffle; test kickoff-named against free options with H54's targets; reconcile the rising work dwell hazard with H129's aging.

- **H138-R1** Move to the call clock.
- **H138-R2** Explain the negative elasticity.
- **H138-R3** Named vs free options.
- **H138-R4** Reconcile dwell.
- **H138-R5** Confirmation.